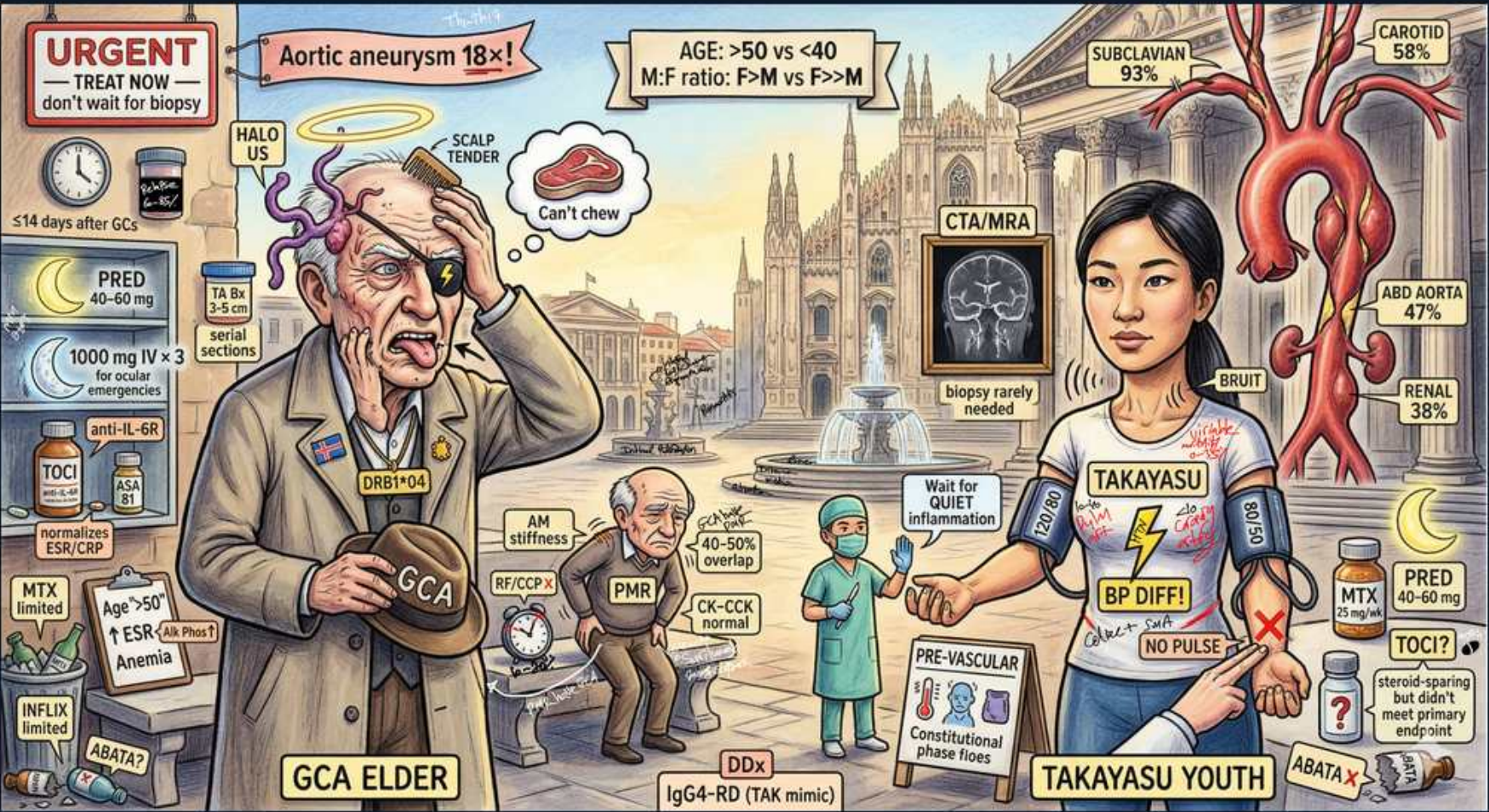


Vasculitis — Large-Vessel: GCA & Takayasu



1. Urgent — treat now

WHAT YOU SEE

URGENT TREAT NOW, don't wait for biopsy banner

WHAT IT ENCODES

GCA is an ophthalmologic emergency: start high-dose glucocorticoids IMMEDIATELY on clinical suspicion to prevent permanent vision loss from arteritic anterior ischemic optic neuropathy. Do not wait for temporal artery biopsy or imaging. Uncomplicated disease: oral prednisone 40-60 mg/day; impending/established vision loss: IV methylprednisolone 1000 mg/day x3 days.

BOARD TRAP

The board trap is choosing 'biopsy first, then treat' — wrong. Visual loss is sudden and irreversible and the fellow eye is at imminent risk; biopsy remains positive for ~1-2 weeks after steroids start, so treating first does not sacrifice the diagnosis.

2. Scalp tender / can't chew

WHAT YOU SEE

Elder holding tender scalp, thought bubble can't chew + meat

WHAT IT ENCODES

GCA cranial symptoms: new temporal headache, scalp tenderness (pain combing hair/on pillow), and jaw claudication — fatigue/pain in the masseters while chewing from ischemia of the muscles of mastication. Tongue claudication and, rarely, tongue/scalp necrosis can occur. Jaw claudication and diplopia are among the strongest predictors of biopsy-positive GCA and of ischemic complications.

BOARD TRAP

Jaw claudication is the MOST specific symptom for GCA — don't dismiss it as TMJ dysfunction or dental pain. Its presence sharply raises the likelihood of GCA and the urgency of steroids; missing it delays sight-saving treatment.

3. Halo on US

WHAT YOU SEE

A HALO US tag pinned over the elder's temple, marking where a temporal-artery Doppler shows the dark 'halo' ring of wall edema around the lumen — the halo IS the inflamed vessel wall.

WHAT IT ENCODES

Temporal/axillary artery color Doppler ultrasound shows the 'halo sign' — a hypoechoic, non-compressible circumferential wall edema around the lumen; a positive compression sign is also characteristic. US is fast, non-invasive, and now a first-line confirmatory test in expert centers, sometimes replacing biopsy. Findings fade within days of steroids, so image early.

BOARD TRAP

A negative ultrasound does NOT exclude GCA, and imaging fades fast once steroids start. Withholding/stopping steroids because the US looks normal is the trap — treat on clinical suspicion and confirm with biopsy if needed.

4. Biopsy <14 days

WHAT YOU SEE

A calendar tag reading '<14 days after GCs' beside the elder — the window during which a temporal-artery biopsy stays positive even after steroids have started.

WHAT IT ENCODES

Temporal artery biopsy is a diagnostic gold standard, showing granulomatous inflammation, giant cells, and fragmentation of the internal elastic lamina. It stays positive for up to ~1-2 weeks (often longer) after steroids begin, so starting treatment does not erase the diagnosis if biopsy is done promptly. Take an adequate length (~1-2 cm) because skip lesions cause false negatives; a contralateral biopsy can be done if the first is negative but suspicion is high.

BOARD TRAP

A NEGATIVE temporal artery biopsy does not rule out GCA (skip lesions, short specimen, wrong side). Stopping steroids in a classic clinical picture because of one negative biopsy is the trap — treat the patient, consider the other side or large-vessel imaging.

5. Pred 40-60 mg

WHAT YOU SEE

A pill bottle labeled PRED 40-60 mg among the elder's meds — the standard oral steroid dose for uncomplicated cranial GCA.

WHAT IT ENCODES

Uncomplicated GCA (no visual or large-vessel ischemic complications): oral prednisone 40-60 mg/day ($\approx$ 1 mg/kg) until symptoms and inflammatory markers normalize, then a slow taper over ~12-24 months. Most patients need bone protection (calcium/vitamin D + bisphosphonate), and low-dose aspirin is sometimes added for ischemic risk. Relapses are common during taper.

BOARD TRAP

Tapering too quickly causes relapse; under-dosing at presentation risks vision loss. Choosing 'low-dose prednisone (e.g., 15 mg)' for cranial GCA is wrong — that PMR-range dose is inadequate for GCA, which needs 40-60 mg (or IV pulse if vision is threatened).

6. 1000 mg IV — vision

WHAT YOU SEE

1000 mg IV x3 / serial sections, for ocular emergencies

WHAT IT ENCODES

Ocular involvement or established/impending vision loss: IV methylprednisolone pulse 1000 mg/day for 3 days, then transition to high-dose oral prednisone. Aggressive treatment aims to salvage remaining vision and protect the fellow eye, which can be lost within days. Tocilizumab is added early as a steroid-sparing agent in relapsing or steroid-toxicity-prone patients.

BOARD TRAP

Visual loss in GCA is usually irreversible, so the goal is protecting the other eye, not reversing established blindness. Reassuring a patient that vision will return with steroids is the trap — and under-treating the threatened eye risks bilateral blindness.

7. Anti-IL-6R (tocilizumab)

WHAT YOU SEE

A vial tagged anti-IL-6R / TOCI with a note that it normalizes ESR/CRP — the IL-6 blockade flattens the inflammatory markers so they can no longer track flares.

WHAT IT ENCODES

Tocilizumab (anti-IL-6 receptor) is FDA-approved, steroid-sparing therapy for GCA that increases sustained glucocorticoid-free remission (GiACTA trial). Caveat: by blocking IL-6 it normalizes ESR and CRP regardless of disease activity, so inflammatory markers can no longer track flares — follow clinical symptoms and imaging instead. Watch for infection, neutropenia, transaminitis, and GI perforation risk.

BOARD TRAP

On tocilizumab, normal ESR/CRP does NOT prove the disease is controlled — IL-6 blockade suppresses these markers artificially. Reassuring or de-escalating based on normalized inflammatory markers in a TCZ-treated patient is the trap; rely on clinical assessment.

8. ESR / CRP high

WHAT YOU SEE

Elevated ESR/CRP and anemia lab tags clustered by the GCA elder — the sky-high sed rate that flags the smoldering arterial inflammation.

WHAT IT ENCODES

GCA typically shows markedly elevated ESR (often >50, classically near 100 mm/hr) and CRP, plus normocytic anemia and thrombocytosis. A very high ESR is a classification criterion. CRP is more sensitive than ESR; checking both increases sensitivity.

BOARD TRAP

A normal ESR does NOT exclude GCA — a small minority have normal ESR, so always check CRP too (and treat on clinical suspicion). Ruling out GCA in a classic vignette because 'ESR is normal' is the trap.

9. PMR link

WHAT YOU SEE

Stooped elder labeled PMR with AM stiffness, RF/CCP negative

WHAT IT ENCODES

Polymyalgia rheumatica overlaps with GCA (up to ~50% of GCA has PMR; ~10-20% of PMR develops GCA): symmetric proximal shoulder- and hip-girdle pain with prominent morning stiffness >45 min, age >50, high ESR/CRP, seronegative (RF/CCP negative), and a dramatic response to LOW-dose prednisone (~15-20 mg/day). CK is normal. Always screen PMR patients for GCA symptoms.

BOARD TRAP

PMR causes muscle PAIN and stiffness with NORMAL CK — it is not myositis (which has weakness + elevated CK). Ordering an EMG/muscle biopsy or starting immunosuppressants for 'myopathy' is the trap. Also: PMR responds to ~15 mg prednisone, whereas concurrent GCA needs 40-60 mg — don't under-treat occult GCA.

10. Aortic aneurysm 18x

WHAT YOU SEE

Aortic aneurysm 18x banner

WHAT IT ENCODES

GCA markedly increases the long-term risk of thoracic aortic aneurysm and dissection (historically cited up to ~17x for thoracic aneurysm), even years after the cranial disease is quiescent. This justifies periodic aortic imaging surveillance (CT/MR/echo) in GCA survivors. Large-vessel GCA can present with limb claudication and bruits, overlapping with Takayasu.

BOARD TRAP

Aortic complications are LATE and can occur after cranial symptoms resolve, so 'GCA is cured once the headache and ESR normalize' is the trap. These patients still need aortic surveillance for delayed aneurysm/dissection.

11. Age >50 vs <40 / F>M

WHAT YOU SEE

AGE >50 vs <40, M/F ratio F>M vs F>>M banner

WHAT IT ENCODES

Age is the primary discriminator between the two large-vessel vasculitides. GCA: age >50 (mean ~70s), female predominant (F>M), more common in those of Northern European descent. Takayasu: age <40-50, strongly female (F>>M), more common in Asian populations. Both can involve the aorta, but cranial/ocular features dominate GCA while aortic-arch branch stenoses dominate Takayasu.

BOARD TRAP

GCA essentially NEVER occurs under age 50 — diagnosing GCA in a 30-year-old woman with subclavian disease is the trap; that is Takayasu. Conversely, large-vessel vasculitis in a 75-year-old is GCA, not new-onset Takayasu.

12. CTA/MRA biopsy rarely

WHAT YOU SEE

CTA/MRA image, biopsy rarely needed, wait for QUIET inflammation

WHAT IT ENCODES

Takayasu is diagnosed by vascular imaging — CTA, MRA, or FDG-PET showing concentric arterial wall thickening, long smooth stenoses, occlusions, and aneurysms of the aorta and its branches; biopsy is rarely needed. Manage active inflammation medically (glucocorticoids ± steroid-sparing agents); defer revascularization (angioplasty/bypass) until inflammation is QUIET to reduce restenosis and complications.

BOARD TRAP

Revascularizing during ACTIVE inflammation is the trap — it leads to high restenosis, anastomotic aneurysm, and graft failure. Suppress inflammation first, then revascularize when the disease is quiet.

13. Takayasu BP diff / no pulse

WHAT YOU SEE

Young woman, BP DIFF, NO PULSE, lightning on arms

WHAT IT ENCODES

Takayasu is 'pulseless disease': inter-arm systolic BP difference (>10-20 mmHg), absent or diminished peripheral pulses, arterial bruits (carotid/subclavian/abdominal), limb claudication, and renovascular hypertension. Constitutional symptoms (fever, weight loss, malaise) often precede vascular findings. Subclavian and carotid stenoses cause arm claudication, dizziness, and visual symptoms.

BOARD TRAP

Measure BP in BOTH arms and use the HIGHER reading (or central/leg pressure) — the stenotic arm underestimates true central pressure. Treating 'normal' BP read from the diseased arm misses dangerous central hypertension; rely on imaging-confirmed pressures.

14. Subclavian/carotid/renal %

WHAT YOU SEE

Aortic tree with SUBCLAVIAN 93%, CAROTID 58%, ABD AORTA 47%, RENAL 38%, BRUIT

WHAT IT ENCODES

Takayasu involves the aorta and major branches in a characteristic distribution: subclavian (most common, ~85-93% — hence arm claudication and pulselessness), carotid (~58% — carotidynia, visual/CNS symptoms), abdominal aorta (~47%), and renal arteries (~38% — renovascular HTN). Bruits over these vessels are common exam findings. The pattern explains the diverse ischemic presentations.

BOARD TRAP

Renovascular hypertension from renal artery stenosis is a major, treatable contributor in Takayasu — attributing the young woman's severe HTN to 'essential hypertension' is the trap. Image the renal arteries and treat the stenosis/inflammation.

15. IgG4-RD mimic

WHAT YOU SEE

DDx IgG4-RD (TAK mimic) tag

WHAT IT ENCODES

IgG4-related disease can cause inflammatory periaortitis/aortitis and retroperitoneal fibrosis that mimics large-vessel vasculitis. Clues to IgG4-RD include elevated serum IgG4, multi-organ involvement (autoimmune pancreatitis, sialadenitis, dacryoadenitis), and storiform fibrosis with IgG4-positive plasma cells on biopsy. It responds to glucocorticoids and rituximab.

BOARD TRAP

Periaortitis is not always Takayasu/GCA — IgG4-RD is a key mimic. Anchoring on 'large-vessel vasculitis' without checking serum IgG4 and looking for pancreatic/glandular disease is the trap; the distinction changes the broader workup and surveillance.